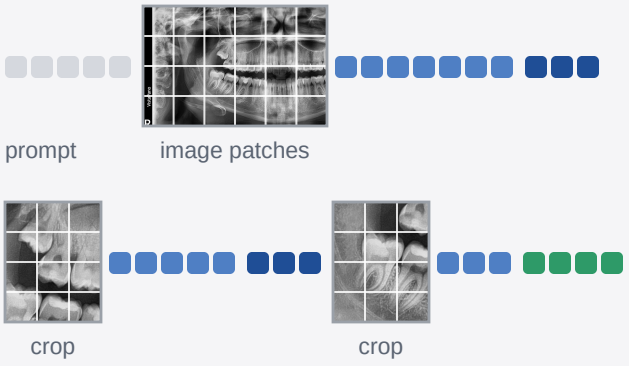
1 SFT on verified traces

1,847 with-tools traces · QLoRA on Qwen3.5-9B

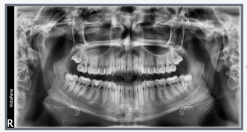
input, masked **loss on:** thought tool call answer



SFT adapter

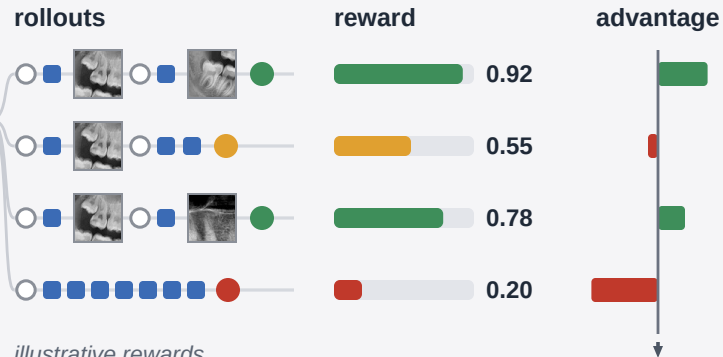
loss on thought, tool call and answer only;
images and tool outputs are inputs, never targets

2 GRPO with live tool rollouts



one image,
K = 4 rollouts

real tools,
same dispatcher



illustrative rewards

A = (r – mean) / (std + ε)

update policy adapter

KL to the frozen SFT adapter

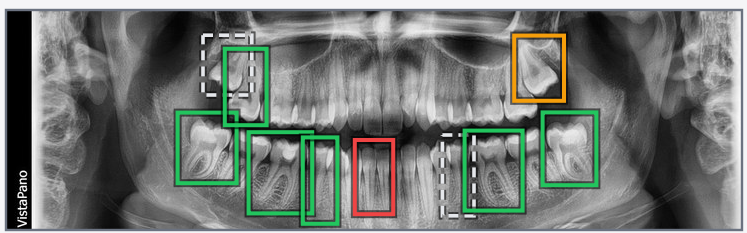
thought tool call tool output crop **answer, reward:** good partial wrong

advantage: above / below the group mean

3 Reward: ground-truth anchored, correction-friendly

R = 1.0 R_acc + 0.2 R_fmt + 0.2 R_tool + 0.1 R_eff

R_acc: matched findings, no LLM judge



exact wrong class missed invented

pair score = 0.25 quadrant + 0.25 tooth + 0.50 diagnosis

image score = F1 of matched recall and precision

R_fmt and R_tool

one valid JSON answer
R_fmt = 1 if it parses

`zoom_crop(bbox)`

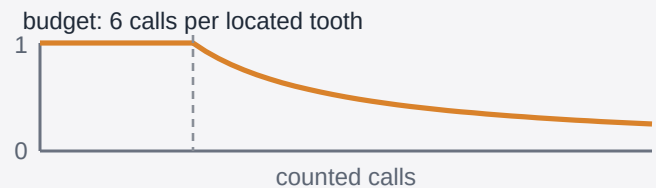
`localize_tooth()`
unknown tool → error

R_tool = valid / attempted calls

R_eff: complexity-scaled, no cost for correction



counted call exempt: locate_tooth, nudge_crop



past the budget: 1 / (1 + overage / budget)